- A particle moving in three-dimensional space. A basis of states for this system is specified by the position of the particle, but this takes three position coordinates. Thus, we have states that are specified by three numbers, $|x, y, z\rangle$. We will see later that all three spatial coordinates of a particle can be simultaneously specified.

- A system composed of two physically independent spins; in other words, a system of two qubits. Later, we will see how to combine systems to form bigger systems. But for now we can just say that the two-spin system can be described by two observables. Namely, we have a state in which both spins are up, another in which both are down, another in which the first is up while the second is down, and another in which these spins are reversed. To put it more briefly, we can characterize the two-spin system by two observables: the z component of the first spin and the z component of the second spin. Quantum mechanics does not forbid simultaneous knowledge of these two observables. In fact, one can choose any component of one spin and any component of the other spin. Quantum mechanics allows simultaneous knowledge of both.

In these situations, we need multiple measurements to fully characterize the state of the system. For example, in our two-spin system, we measure each spin separately and associate these measurements with two different operators. We'll call these operators **L** and **M**.

A measurement leaves the system in an eigenstate (consisting of a single eigenvector), corresponding to the value (an eigenvalue) that was measured. If we measure both spins in a two-spin system, the system winds up in a state that is simultaneously an eigenvector of **L** and an eigenvector of **M**. We call this a *simultaneous eigenvector* of the operators **L** and **M**.

The two-spin example gives us something concrete to think about, but keep in mind that our results will be far more general—they will apply to any system that is characterized by two different operators. And as you